

AI HOME DESIGNER
FRONTEND SPECIFICATION
PART 1 + PART 2

STATUS:
APPLICATION ARCHITECTURE + DESIGN SYSTEM

=====

=====

1. FRONTEND STACK

=====

=====

FRAMEWORK

Next.js

LANGUAGE

TypeScript

UI

React

STYLING

TailwindCSS

STATE

Zustand

FORMS

React Hook Form

CANVAS

Konva.js

3D

Three.js

=====

=====

2. APPLICATION STRUCTURE

=====

=====

app/

components/

features/

hooks/

services/

stores/

types/

utils/

styles/

assets/

=====

=====

3. GLOBAL LAYOUT

=====

=====

Top Navigation

↓

Left Sidebar

↓

Center Canvas

↓

Right Properties Panel

↓

Bottom Status Bar

=====

4. MAIN PAGES

=====

Login

Register

Dashboard

Projects

Editor

AI Assistant

Templates

Materials

Profile

Settings

Billing

Admin

=====

5. DESIGN SYSTEM

=====

Spacing

8px Grid

Border Radius

4

8

12

16

Typography

Display

Heading

Body

Caption

=====

=====

6. COLOR TOKENS

=====

=====

Primary

Secondary

Success

Warning

Danger

Background

Surface

Border

Text

Muted

Dark Mode Supported

=====

=====

7. ICON SYSTEM

=====

=====

Icons For

Rooms

Walls

Doors

Windows

Roof

Furniture

Export

AI

Settings

Notifications

=====

=====

8. RESPONSIVE BREAKPOINTS

=====

=====

Mobile

<640px

Tablet

640–1024px

Desktop

1024–1536px

Ultra Wide

>1536px

=====

9. NAVIGATION

=====

Dashboard

Projects

Templates

AI

Materials

Settings

Profile

=====

10. PROJECT DASHBOARD

=====

Widgets

Recent Projects

Favorites

Recent AI Jobs

Notifications

Storage Usage

Statistics

```
=====
=====
11. PROJECT CARD
=====
=====
```

Displays

Thumbnail

Project Name

Owner

Area

Rooms

Last Modified

Status

Favorite

```
=====
=====
12. SEARCH & FILTER
=====
=====
```

Search

Project Type

Tags

Status

Date

Owner

Favorites

=====

=====

13. USER MENU

=====

=====

Profile

Preferences

Billing

API Keys

Logout

=====

=====

14. APPLICATION STATUS

=====

=====

Top Right Indicators

Online

Offline

Saving

Syncing

AI Running

Export Running

=====

=====

15. NEXT

=====

=====

Editor

Canvas

Toolbar

Property Panel

Keyboard Shortcuts

=====

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END OF FRONTEND SPECIFICATION PARTS 1 + 2

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=====

AI HOME DESIGNER
FRONTEND SPECIFICATION
PART 3 + PART 4

STATUS:
2D EDITOR + CANVAS + TOOLBAR + PROPERTY PANEL

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=====

16. EDITOR LAYOUT

=====

=====

MAIN STRUCTURE

Top Toolbar

↓

Left Tool Palette

↓

Center Infinite Canvas

↓

Right Property Panel

↓

Bottom Status Bar

Resizable Panels

Collapsible Sidebar

Fullscreen Mode

=====

=====

17. CANVAS FEATURES

=====

=====

Infinite Canvas

Zoom In / Out

Pan

Mini Map

Grid

Snap to Grid

Snap to Objects

Multi Select

Undo / Redo

Auto Save

=====

=====

18. DRAWING TOOLS

=====

=====

Select

Move

Room

Wall

Door

Window

Roof

Stair

Furniture

Measure

Text

Delete

=====

=====

19. ROOM EDITOR

=====

=====

FUNCTIONS

Create Room

Resize

Move

Rotate

Rename

Duplicate

Delete

Merge

Split

Change Type

=====

=====

20. WALL EDITOR

=====

=====

FUNCTIONS

Draw Wall

Move Wall

Extend

Trim

Split

Merge

Change Thickness

Change Material

=====

=====

21. DOOR & WINDOW EDITOR

=====

=====

FUNCTIONS

Insert

Move

Resize

Flip Direction

Change Type

Replace

Delete

=====

=====

22. PROPERTY PANEL

=====

=====

Displays Selected Object

GENERAL

ID

Name

Type

GEOMETRY

Position

Width

Height

Rotation

MATERIALS

Material

Finish

Cost

ADVANCED

Tags

Metadata

AI Notes

=====

=====

23. TOOLBAR

=====

=====

New Project

Open

Save

Undo

Redo

Copy

Paste

Delete

AI Generate

AI Optimize

Validate

Export

Settings

=====

=====

24. KEYBOARD SHORTCUTS

=====

=====

Ctrl + N

New Project

Ctrl + S

Save

Ctrl + Z

Undo

Ctrl + Y

Redo

Ctrl + C

Copy

Ctrl + V

Paste

Ctrl + D

Duplicate

Delete

Remove Object

Space

Pan Canvas

Mouse Wheel

Zoom

=====

=====

25. OBJECT SELECTION

=====

=====

Single Select

Multi Select

Selection Box

Select All

Invert Selection

Lock Selection

Hide Selection

=====

=====

26. GRID SYSTEM

=====

=====

Grid Sizes

5 cm

10 cm

25 cm

50 cm

100 cm

Snap

ON/OFF

Dynamic Grid

=====

=====

27. STATUS BAR

=====

=====

Shows

Mouse Position

Zoom Level

Selected Objects

Project Status

Auto Save Status

Validation Status

AI Status

=====

=====

28. CONTEXT MENU

=====

=====

Right Click

Copy

Paste

Duplicate

Delete

Lock

Hide

Bring Forward

Send Backward

Properties

=====

=====

29. EDITOR PERFORMANCE

=====

=====

Target

60 FPS

100,000 Objects Supported

Incremental Rendering

Virtualized Object List

Canvas Caching

=====

=====

30. NEXT

=====

=====

3D Viewer

AI Assistant

Material Library

Template System

Collaboration

```
=====
=====
END OF FRONTEND SPECIFICATION PARTS 3 + 4
=====
=====
```

AI HOME DESIGNER
FRONTEND SPECIFICATION
PART 5 + PART 6

STATUS:
3D VIEWER + AI ASSISTANT + MATERIAL LIBRARY + TEMPLATES

```
=====
=====
31. 3D VIEWER
=====
=====
```

FEATURES

3D Navigation

Orbit Camera

Walk Mode

First Person

Top View

Section View

Exploded View

Fullscreen

Screenshot

```
=====
=====
32. CAMERA CONTROLS
=====
=====
```

Rotate

Pan

Zoom

Reset View

Focus Selected Object

Save Camera Position

Restore Camera Position

=====

33. RENDER OPTIONS

=====

Wireframe

Solid

Realistic

Shadows

Ambient Occlusion

Reflections

Transparency

Edge Display

=====

34. AI ASSISTANT PANEL

=====

FEATURES

Natural Language Chat

Generate Floor Plan

Optimize Layout

Explain Design

Suggest Improvements

Estimate Cost

Answer Building Questions

=====

=====

35. AI PROMPT INPUT

=====

=====

FIELDS

Project Description

House Type

Number of Floors

Target Area

Budget

Style

Special Requirements

Local Building Rules

=====

=====

36. AI RESULT PANEL

=====

=====

DISPLAYS

Generated Layout

Score

Warnings

Suggestions

Estimated Cost

Energy Rating

Accept

Reject

Regenerate

=====

=====

37. MATERIAL LIBRARY

=====

=====

CATEGORIES

Walls

Floors

Roof

Doors

Windows

Paint

Tiles

Wood

Steel

Concrete

Glass

Insulation

=====

=====

38. MATERIAL DETAILS

=====

SHOWS

Image

Name

Manufacturer

Price

Thermal Value

Fire Rating

Sound Rating

Carbon Footprint

Availability

=====

39. TEMPLATE LIBRARY

=====

TEMPLATE TYPES

Tiny House

Family House

Villa

Apartment

Office

Warehouse

Commercial

Restaurant

Custom

=====

=====

40. TEMPLATE ACTIONS

=====

=====

Preview

Duplicate

Favorite

Apply

Share

Delete

Create New

=====

=====

41. TEMPLATE FILTERS

=====

=====

Area

Bedrooms

Bathrooms

Floors

Style

Budget

Country

Energy Class

=====

=====

42. MATERIAL SEARCH

=====

=====

FILTERS

Category

Brand

Price

Energy Rating

Fire Rating

Availability

=====

43. RECENT ITEMS

=====

Displays

Recent Templates

Recent Materials

Recent AI Prompts

Recent Projects

=====

44. QUICK ACTIONS

=====

Generate House

Optimize Project

Validate Design

Estimate Cost

Export PDF

Export IFC

Share Project

```
=====
=====
45. NEXT
=====
=====
```

Real-time Collaboration

Comments

Version Comparison

Presentation Mode

Mobile UI

```
=====
=====
END OF FRONTEND SPECIFICATION PARTS 5 + 6
=====
=====
```

AI HOME DESIGNER
FRONTEND SPECIFICATION
PART 7 + PART 8

STATUS:
REAL-TIME COLLABORATION + VERSIONING + MOBILE UI

```
=====
=====
46. REAL-TIME COLLABORATION
=====
=====
```

FEATURES

Multi-user Editing

Live Cursor

Live Object Selection

Live Presence

Conflict Detection

Auto Synchronization

Session Recovery

```
=====
=====
47. USER PRESENCE
=====
=====
```

DISPLAY

User Name

Avatar

Cursor Position

Selected Object

Current Tool

Current Floor

Editing Status

```
=====
=====
48. COMMENT SYSTEM
=====
=====
```

FEATURES

Create Comment

Reply

Resolve

Reopen

Mention User

Attach Screenshot

Attach Object

Filter Comments

=====

=====

49. VERSION COMPARISON

=====

=====

COMPARE

Room Changes

Wall Changes

Door Changes

Window Changes

Material Changes

Cost Changes

AI Changes

=====

=====

50. VERSION TIMELINE

=====

=====

ACTIONS

Create Version

Restore Version

Rename Version

Delete Version

Compare Versions

Export Version

Lock Version

=====

=====

51. PRESENTATION MODE

=====

=====

FEATURES

Fullscreen

Hide Editing Tools

Room Information

Interactive Walkthrough

Slideshow

Client View

QR Code Sharing

=====

=====

52. MOBILE APPLICATION

=====

=====

SCREENS

Dashboard

Projects

Viewer

Comments

Notifications

Profile

Settings

EDITOR

Read-only by default

Basic Editing Optional

=====

=====

53. TABLET MODE

=====

=====

FEATURES

Full 2D Editor

Touch Gestures

Apple Pencil Support

Stylus Support

Split View

Landscape Optimization

=====

=====

54. OFFLINE MODE

=====

=====

FEATURES

Local Cache

Offline Editing

Auto Sync

Conflict Resolution

Offline Export Queue

=====

=====

55. ACCESSIBILITY

=====

=====

SUPPORT

Keyboard Navigation

Screen Readers

High Contrast

Large Text

Reduced Motion

Color Blind Friendly

=====

56. LOCALIZATION

=====

SUPPORTED LANGUAGES

English

Romanian

Hungarian

German

French

Spanish

Italian

Polish

Expandable

=====

57. PERFORMANCE TARGETS

=====

Application Load

<2 seconds

Project Open

<1 second

Canvas FPS

60 FPS

Auto Save

<300 ms

AI Preview

<5 seconds

=====

=====

58. FRONTEND SECURITY

=====

=====

FEATURES

JWT Storage

CSRF Protection

XSS Protection

CSP

Secure Cookies

Content Sanitization

=====

=====

59. FRONTEND STATUS

=====

=====

UI SPECIFICATION

100% COMPLETE

Production Ready

=====

=====

60. NEXT DOCUMENT

=====

=====

AI PROMPT LIBRARY

Contents

- System Prompt
- Planner Prompt
- Validator Prompt
- Optimizer Prompt
- Cost Estimator Prompt
- Energy Analyzer Prompt
- Interior Designer Prompt
- Prompt Templates

=====

=====

END OF FRONTEND SPECIFICATION

FRONTEND DESIGN COMPLETE

=====

=====

AI HOME DESIGNER

AI PROMPT LIBRARY

PART 1 + PART 2

STATUS:

SYSTEM PROMPTS + FLOORPLAN GENERATION

=====

=====

1. MASTER SYSTEM PROMPT

=====

=====

ROLE

You are an expert architectural AI specialized in residential and commercial building design.

PRIMARY OBJECTIVES

- Generate valid floor plans
- Respect building constraints
- Optimize functionality
- Minimize construction cost
- Maximize usability
- Produce deterministic JSON output

NEVER

- Invent unsupported objects
- Return invalid geometry
- Ignore mandatory requirements
- Produce explanations instead of structured output

=====

=====

2. OUTPUT FORMAT

=====

=====

ALWAYS RETURN

JSON ONLY

NO Markdown

NO Comments

NO Additional Text

Schema validation required.

=====

=====

3. FLOORPLAN GENERATION PROMPT

=====

=====

INPUT

Building Type

Total Area

Number of Floors

Room Requirements

Building Style

Budget

Country

Climate Zone

Building Regulations

OUTPUT

Complete layout JSON

```
=====
=====
4. ROOM PLANNING RULES
=====
=====
```

PRIORITY

1

Functional circulation

2

Natural lighting

3

Privacy

4

Construction simplicity

5

Future expansion

=====

5. ROOM RELATIONSHIPS

=====

Kitchen

Near Dining

Dining

Near Living Room

Bathroom

Accessible

Garage

Near Entrance

Laundry

Near Utility

=====

6. HALLWAY OPTIMIZATION

=====

RULE

Minimize hallway area.

Target

<10%

of total usable floor area.

=====

7. VALID ROOM SIZES

=====

AI must respect

Minimum Width

Minimum Height

Minimum Area

Maximum Area

Accessibility Rules

=====

8. WALL GENERATION

=====

RULES

Closed Geometry

No Gaps

No Overlaps

Connected Structure

Consistent Thickness

=====

=====

9. DOOR PLACEMENT

=====

=====

RULES

Accessible

No Collision

Correct Swing Direction

Minimum Clearance

Logical Room Flow

=====

=====

10. WINDOW PLACEMENT

=====

=====

RULES

Natural Light

Cross Ventilation

Energy Efficiency

Exterior Walls Only

=====

=====

11. GENERATION PRIORITIES

=====

=====

Priority Order

1

Safety

2

Geometry

3

Building Rules

4

Functionality

5

Comfort

6

Energy

7

Cost

=====
=====

12. AI FAILURE POLICY

=====
=====

If requirements conflict

↓

Return validation errors

Do not invent impossible layouts.

=====
=====

13. NEXT

=====
=====

Validator Prompt

Optimization Prompt

Cost Estimation Prompt

=====

=====

END OF AI PROMPT LIBRARY PARTS 1 + 2

=====

=====

AI HOME DESIGNER
AI PROMPT LIBRARY
PART 3 + PART 4

STATUS:
VALIDATOR + OPTIMIZER + COST ESTIMATOR

=====

=====

14. VALIDATOR SYSTEM PROMPT

=====

=====

ROLE

You are a professional architectural validation engine.

Your task is NOT to redesign the project.

Your task is ONLY to verify whether the submitted design
complies with all architectural, structural and application rules.

Return a structured JSON validation report.

=====

=====

15. VALIDATION CHECKS

=====

=====

GEOMETRY

- ✓ Closed rooms
- ✓ Connected walls
- ✓ No overlaps
- ✓ No duplicated objects

- ✓ Valid coordinates

ROOMS

- ✓ Minimum dimensions
- ✓ Correct room types
- ✓ Logical placement

DOORS

- ✓ Reachable
- ✓ Correct opening
- ✓ No collision

WINDOWS

- ✓ Exterior wall
- ✓ Proper daylight
- ✓ Ventilation

BUILDING

- ✓ Total area
- ✓ Floor count
- ✓ Accessibility

16. VALIDATION RESPONSE

RETURN

Overall Score

Geometry Score

Accessibility Score

Energy Score

Construction Score

Cost Score

Also return

Errors[]

Warnings[]

Suggestions[]

=====

=====

17. OPTIMIZER SYSTEM PROMPT

=====

=====

ROLE

Improve an existing design.

Do NOT redesign from scratch.

Only make measurable improvements.

=====

=====

18. OPTIMIZATION TARGETS

=====

=====

Reduce corridor area

Improve room flow

Reduce wall length

Lower construction cost

Increase natural lighting

Improve ventilation

Improve furniture placement

Increase energy efficiency

=====

=====

19. OPTIMIZATION LIMITS

=====

=====

DO NOT

Remove mandatory rooms

Break geometry

Violate building rules

Increase cost unnecessarily

Change user-defined constraints

=====

=====

20. COST ESTIMATOR PROMPT

=====

=====

ROLE

Estimate realistic construction costs using
the current material database and regional pricing.

=====

=====

21. COST BREAKDOWN

=====

=====

RETURN

Foundation

Walls

Roof

Windows

Doors

Flooring

Electrical

Plumbing

HVAC

Interior Finish

Labor

Equipment

Transport

Taxes

Contingency

TOTAL COST

Cost per m²

=====

=====

22. ENERGY ANALYZER PROMPT

=====

=====

ANALYZE

Heat Loss

Solar Gain

Window Efficiency

Roof Efficiency

Wall Insulation

Heating Demand

Cooling Demand

Annual Energy Consumption

Expected Energy Class

=====

23. INTERIOR DESIGN PROMPT

=====

ROLE

Automatically furnish rooms while respecting:

Clear walking paths

Door swing clearance

Window accessibility

Furniture ergonomics

Style consistency

Budget

=====

24. AI EXPLANATION PROMPT

=====

When requested, explain:

Why a room was placed

Why a wall moved

Why a warning appeared

Why optimization was suggested

Use concise, technically accurate explanations.

=====

=====

25. PROMPT EXECUTION ORDER

=====

=====

1. Planner

↓

2. Validator

↓

3. Optimizer

↓

4. Validator

↓

5. Cost Estimator

↓

6. Energy Analyzer

↓

7. Interior Designer (optional)

=====

=====

26. AI LIBRARY STATUS

=====

=====

Prompt Library

100% Complete

Production Ready

=====
=====
27. NEXT DOCUMENT
=====
=====

JSON Schema Library

Contents

- Project Schema
- Room Schema
- Wall Schema
- Door Schema
- Window Schema
- Roof Schema
- Material Schema
- Validation Schemas

=====
=====
END OF AI PROMPT LIBRARY
AI PROMPT DESIGN COMPLETE
=====
=====

AI HOME DESIGNER
JSON SCHEMA LIBRARY
PART 1 + PART 2

STATUS:
CORE DATA STRUCTURES

=====
=====
1. ROOT PROJECT OBJECT

```
=====
=====

{
  "schema_version": "1.0",
  "project": {},
  "floors": [],
  "rooms": [],
  "walls": [],
  "doors": [],
  "windows": [],
  "roofs": [],
  "stairs": [],
  "materials": [],
  "furniture": [],
  "electrical": [],
  "plumbing": [],
  "hvac": [],
  "metadata": {}
}
```

```
=====
=====
2. PROJECT SCHEMA
=====
=====
```

```
{
  "id": "UUID",
  "name": "string",
  "owner_id": "UUID",
  "project_type": "HOUSE",
  "country": "RO",
  "city": "string",
  "site_area": 0,
  "building_area": 0,
  "floors": 1,
  "status": "DRAFT"
}
```

```
=====
=====
3. FLOOR SCHEMA
=====
=====
```

```
{
  "id": "UUID",
```

```
"level": 0,  
"name": "Ground Floor",  
"height": 2.80,  
"usable_area": 0,  
"rooms": []  
}
```

```
=====
```

```
=====
```

```
4. ROOM SCHEMA
```

```
=====
```

```
=====
```

```
{  
  "id": "UUID",  
  "type": "BEDROOM",  
  "name": "Bedroom 1",  
  "floor": 0,  
  "x": 0,  
  "y": 0,  
  "width": 400,  
  "height": 350,  
  "area": 14.0,  
  "rotation": 0  
}
```

```
=====
```

```
=====
```

```
5. WALL SCHEMA
```

```
=====
```

```
=====
```

```
{  
  "id": "UUID",  
  "room_id": "UUID",  
  "start": { "x": 0, "y": 0 },  
  "end": { "x": 400, "y": 0 },  
  "height": 280,  
  "thickness": 30,  
  "material": "brick"  
}
```

```
=====
```

```
=====
```

```
6. DOOR SCHEMA
```

```
=====
```

```
=====
```



```
{
  "id": "UUID",
  "wall_id": "UUID",
  "type": "HINGED",
  "width": 90,
  "height": 210,
  "position": 120,
  "direction": "LEFT"
}
```

```
=====
=====
```

7. WINDOW SCHEMA

```
=====
=====
```

```
{
  "id": "UUID",
  "wall_id": "UUID",
  "type": "CASEMENT",
  "width": 150,
  "height": 120,
  "sill_height": 90,
  "glass_layers": 3
}
```

```
=====
=====
```

8. ROOF SCHEMA

```
=====
=====
```

```
{
  "id": "UUID",
  "type": "GABLE",
  "pitch": 35,
  "material": "tile",
  "area": 180
}
```

```
=====
=====
```

9. STAIR SCHEMA

```
=====
=====
```

```
{
  "id": "UUID",
```

```
"type": "L_SHAPED",
"steps": 18,
"width": 100,
"rise": 17,
"run": 28
}
```

```
=====
=====
10. MATERIAL SCHEMA
=====
=====
```

```
{
  "id": "UUID",
  "category": "WOOD",
  "name": "Oak",
  "unit": "m²",
  "price": 52.5,
  "currency": "EUR"
}
```

```
=====
=====
11. METADATA
=====
=====
```

```
{
  "created_at": "",
  "updated_at": "",
  "version": 1,
  "author": "",
  "generator": "AI"
}
```

```
=====
=====
12. NEXT
=====
=====
```

Validation Schemas

API Payloads

Import/Export Schemas

```
=====
=====
END OF JSON SCHEMA LIBRARY PARTS 1 + 2
=====
=====
```

```
# AI HOME DESIGNER
# JSON SCHEMA LIBRARY
# PART 3 + PART 4
```

```
STATUS:
VALIDATION SCHEMAS + IMPORT/EXPORT + AI PAYLOADS
```

```
=====
=====
13. VALIDATION RESULT SCHEMA
=====
=====
```

```
{
  "success": true,
  "overall_score": 96,
  "geometry_score": 98,
  "building_score": 95,
  "energy_score": 93,
  "cost_score": 91,
  "accessibility_score": 97,
  "errors": [],
  "warnings": [],
  "suggestions": []
}
```

```
=====
=====
14. VALIDATION ERROR SCHEMA
=====
=====
```

```
{
  "code": "ROOM_TOO_SMALL",
  "severity": "ERROR",
  "object_type": "ROOM",
  "object_id": "UUID",
  "message": "",
  "recommendation": ""
}
```

```
-----
```

SUPPORTED SEVERITY

INFO

WARNING

ERROR

CRITICAL

```
=====
=====
15. AI GENERATION REQUEST
=====
=====
```

```
{
  "project_id": "UUID",
  "version_id": "UUID",
  "prompt": "",
  "building_type": "HOUSE",
  "floors": 2,
  "target_area": 180,
  "budget": 250000,
  "country": "RO",
  "style": "Modern"
}
```

```
=====
=====
16. AI GENERATION RESPONSE
=====
=====
```

```
{
  "generation_id": "UUID",
  "status": "SUCCESS",
  "layout": {},
  "validation": {},
  "cost": {},
  "energy": {},
  "processing_time_ms": 4250
}
```

```
=====
=====
17. EXPORT REQUEST
```

```
=====
=====

{
  "project_id": "UUID",
  "format": "PDF",
  "include_costs": true,
  "include_materials": true,
  "include_ai_notes": false,
  "include_3d": true
}
```

```
=====
=====
18. EXPORT RESPONSE
=====
=====
```

```
{
  "export_id": "UUID",
  "status": "COMPLETED",
  "download_url": "",
  "file_size": 8421562,
  "expires_at": ""
}
```

```
=====
=====
19. IMPORT SCHEMA
=====
=====
```

```
{
  "schema_version": "1.0",
  "source": "DXF",
  "project": {},
  "objects": [],
  "metadata": {}
}
```

```
=====
=====
20. BULK UPDATE SCHEMA
=====
=====
```

```
{
  "operation": "UPDATE",
```

```
"objects": [  
  {  
    "id": "UUID",  
    "changes": {}  
  }  
]  
}
```

```
=====
```

21. EVENT PAYLOAD

```
=====
```

```
{  
  "event": "project.updated",  
  "timestamp": "",  
  "user_id": "UUID",  
  "project_id": "UUID",  
  "payload": {}  
}
```

```
=====
```

22. AUDIT ENTRY

```
=====
```

```
{  
  "action": "UPDATE",  
  "entity": "ROOM",  
  "entity_id": "UUID",  
  "before": {},  
  "after": {},  
  "performed_by": "UUID",  
  "created_at": ""  
}
```

```
=====
```

23. SCHEMA VERSIONING

```
=====
```

RULES

Major Version

Breaking Changes

Minor Version

Backward Compatible

Patch

Bug Fixes

=====

=====

24. COMPLETE JSON LIBRARY

=====

=====

STATUS

Project Schema ✓

Room Schema ✓

Wall Schema ✓

Door Schema ✓

Window Schema ✓

Roof Schema ✓

Material Schema ✓

Validation Schema ✓

Import/Export ✓

AI Payloads ✓

Production Ready

=====

=====

25. NEXT DOCUMENT

=====

AI ENGINE SPECIFICATION

Contents

- Multi-Agent Architecture
- Planning Engine
- Geometry Engine
- Rule Engine
- Optimization Engine
- Vision Engine
- LLM Orchestration
- Memory System

=====

END OF JSON SCHEMA LIBRARY
JSON SPECIFICATION COMPLETE

=====

AI HOME DESIGNER
AI ENGINE SPECIFICATION
PART 1 + PART 2

STATUS:
MULTI-AGENT AI ARCHITECTURE

=====

1. AI ENGINE OVERVIEW

=====

The AI system is NOT a single model.

It is an orchestrated Multi-Agent Architecture where specialized AI agents cooperate to generate, validate, optimize and explain architectural designs.

=====

=====

2. AI AGENTS

=====

=====

MASTER ORCHESTRATOR

↓

Planner Agent

↓

Geometry Agent

↓

Building Rules Agent

↓

Cost Agent

↓

Energy Agent

↓

Interior Agent

↓

Validator Agent

↓

Explanation Agent

=====

=====

3. MASTER ORCHESTRATOR

=====

=====

RESPONSIBILITIES

Receive user request

Select required agents

Execute workflow

Merge outputs

Resolve conflicts

Return final project

=====

=====

4. PLANNER AGENT

=====

=====

RESPONSIBILITIES

Interpret user intent

Determine required rooms

Create adjacency graph

Estimate dimensions

Generate first layout proposal

=====

=====

5. GEOMETRY AGENT

=====

=====

RESPONSIBILITIES

Generate coordinates

Create walls

Place doors

Place windows

Generate roof

Generate stairs

Maintain valid geometry

=====

6. BUILDING RULE AGENT

=====

RESPONSIBILITIES

Apply national regulations

Minimum room sizes

Accessibility

Fire safety

Emergency exits

Structural constraints

=====

7. COST AGENT

=====

RESPONSIBILITIES

Estimate material quantities

Estimate labor

Estimate transport

Calculate taxes

Generate construction budget

Optimize cost

=====

8. ENERGY AGENT

=====

RESPONSIBILITIES

Thermal analysis

Solar analysis

Ventilation

Heat loss

Cooling demand

Energy certificate estimation

=====

9. INTERIOR AGENT

=====

RESPONSIBILITIES

Furniture placement

Walking paths

Kitchen layout

Bathroom layout

Storage optimization

Style consistency

=====

10. VALIDATOR AGENT

=====

RESPONSIBILITIES

Validate geometry

Validate regulations

Validate accessibility

Validate AI output

Return structured errors

=====

=====

11. EXPLANATION AGENT

=====

=====

RESPONSIBILITIES

Explain decisions

Generate summaries

Answer user questions

Explain warnings

Explain optimization

=====

=====

12. MEMORY MANAGER

=====

=====

Stores

Conversation

Project History

AI Decisions

Accepted Designs

Rejected Designs

Optimization History

=====

=====

13. EXECUTION PIPELINE

=====

User Request



Planner



Geometry



Rules



Validator



Optimizer



Validator



Cost



Energy



Interior



Explanation



Final JSON

=====

=====

14. FAILURE HANDLING

=====

=====

If one agent fails

↓

Retry

↓

Fallback Agent

↓

Partial Recovery

↓

Return Structured Error

=====

=====

15. NEXT

=====

=====

Geometry Engine

Constraint Solver

Optimization Algorithms

Learning System

=====

=====

END OF AI ENGINE SPECIFICATION PARTS 1 + 2

=====

=====

AI HOME DESIGNER
AI ENGINE SPECIFICATION
PART 3 + PART 4

STATUS:
GEOMETRY ENGINE + CONSTRAINT SOLVER + OPTIMIZATION ENGINE

=====

=====

16. GEOMETRY ENGINE

=====

=====

PURPOSE

Generate mathematically valid building geometry.

INPUT

Project Requirements

↓

Floor Definitions

↓

Room Graph

↓

Building Rules

OUTPUT

Complete Parametric Geometry

=====

=====

17. GEOMETRY PIPELINE

=====

=====

Step 1

Generate Building Envelope

↓

Step 2

Split Floors



Step 3

Generate Rooms



Step 4

Generate Walls



Step 5

Insert Doors



Step 6

Insert Windows



Step 7

Generate Roof



Step 8

Validation

=====

=====

18. CONSTRAINT SOLVER

=====

=====

Every generated object must satisfy all active constraints.

Constraint Categories

- Geometric
- Architectural
- Structural
- Accessibility
- HVAC
- Plumbing
- Electrical
- User-defined

=====

=====

19. HARD CONSTRAINTS

=====

=====

Cannot be violated.

Examples

- ✓ No overlapping rooms
- ✓ Closed wall loops
- ✓ Doors inside walls
- ✓ Windows only on valid walls
- ✓ Minimum room dimensions
- ✓ Valid stair geometry

=====

=====

20. SOFT CONSTRAINTS

=====

=====

Optimization targets.

Examples

- Shorter corridors
- Better sunlight
- Lower cost
- Less wall length

- Better furniture flow
- Better energy performance

=====

=====

21. OBJECTIVE FUNCTION

=====

=====

Overall Design Score

=

Geometry

+

Functionality

+

Energy

+

Cost

+

Comfort

+

Expandability

+

Maintainability

Weighted score

0–100

=====

=====

22. OPTIMIZATION ENGINE

=====

=====

ALGORITHMS

Rule-based Optimization

Constraint Satisfaction

Genetic Algorithms (optional)

Simulated Annealing

Beam Search

Greedy Improvements

LLM-guided Refinement

=====

=====

23. ITERATIVE IMPROVEMENT

=====

=====

Initial Layout

↓

Validate

↓

Optimize

↓

Validate Again

↓

Repeat Until

- Score threshold reached

OR

- Iteration limit reached

=====

=====

24. GEOMETRY VALIDATION

=====

=====

CHECKS

Closed polygons

No intersections

Correct wall orientation

Connected graph

Area consistency

Volume consistency

Coordinate precision

=====

=====

25. PATHFINDING ENGINE

=====

=====

PURPOSE

Verify movement through the building.

CHECKS

Entrance → Living Area

Living → Kitchen

Bedroom → Bathroom

Emergency Exit Routes

Wheelchair Routes

=====

=====

26. COLLISION ENGINE

=====

Detect

Furniture collisions

Door swing conflicts

Window conflicts

Wall overlaps

Stair conflicts

Roof intersections

=====

27. AUTO-CORRECTION ENGINE

=====

Automatically fixes

Minor wall gaps

Coordinate rounding

Small overlaps

Door alignment

Window alignment

Grid snapping

Never changes user-defined mandatory requirements.

=====

28. PERFORMANCE TARGETS

=====

Typical House

<5 seconds

Large Villa

<10 seconds

Apartment Building

<30 seconds

Commercial Building

<60 seconds

=====

=====

29. AI ENGINE STATUS

=====

=====

Multi-Agent System ✓

Constraint Solver ✓

Geometry Engine ✓

Optimization Engine ✓

Production Ready

=====

=====

30. NEXT DOCUMENT

=====

=====

3D Engine Specification

- Rendering Engine
- IFC Generator
- DXF Generator
- GLTF Export
- Ray Casting
- Camera System
- Material System
- Lighting Engine

=====

=====

END OF AI ENGINE SPECIFICATION PARTS 3 + 4

AI ENGINE CORE COMPLETE

=====

=====

AI HOME DESIGNER

3D ENGINE SPECIFICATION

PART 1 + PART 2

STATUS:

3D RENDERING + MODEL GENERATION

=====

=====

1. 3D ENGINE OVERVIEW

=====

=====

PURPOSE

Convert every validated 2D project into a complete parametric 3D building.

INPUT

Project JSON

↓

Geometry

↓

Materials

↓

Roof

↓

Furniture

↓

Lighting

OUTPUT

Interactive 3D Scene

=====

=====

2. RENDERING ENGINE

=====

=====

ENGINE

Three.js

FEATURES

Real-Time Rendering

GPU Acceleration

PBR Materials

HDR Support

Shadows

Reflections

Post Processing

=====

=====

3. MODEL GENERATION PIPELINE

=====

=====

Project JSON

↓

Walls

↓

Doors

↓

Windows

↓

Floors

↓

Roof

↓

Furniture

↓

Lights

↓

Scene Graph

=====

4. WALL MESH GENERATOR

=====

GENERATES

Wall Mesh

Inner Faces

Outer Faces

Top Faces

Material Layers

Wall Openings

=====

5. DOOR GENERATOR

=====

GENERATES

Door Frame

Door Leaf

Handle

Hinges

Opening Animation

Collision Box

=====

6. WINDOW GENERATOR

=====

GENERATES

Frame

Glass

Sill

Trim

Opening Animation

Reflection Material

=====

7. ROOF GENERATOR

=====

GENERATES

Roof Surface

Ridge

Eaves

Gutters

Fascia

Insulation Layers

=====

8. FLOOR GENERATOR

=====

GENERATES

Concrete Slab

Floor Finish

Insulation

Base Layer

Ceiling Below

=====

9. MATERIAL SYSTEM

=====

SUPPORTED

PBR Materials

Normal Maps

Roughness Maps

Metallic Maps

AO Maps

Transparency

Emissive Materials

=====

=====

10. LIGHTING ENGINE

=====

=====

LIGHT TYPES

Directional

Point

Spot

Area

HDR Environment

Sun Simulation

Artificial Lights

=====

=====

11. CAMERA SYSTEM

=====

=====

MODES

Orbit

Walk

First Person

Top View

Section View

Isometric

Free Camera

=====

=====

12. INTERACTION

=====

SUPPORTED

Object Selection

Measure Distance

Hide Objects

Explode Model

Cut Sections

Annotations

=====

13. PERFORMANCE TARGETS

=====

Target FPS

60 FPS

Scene Size

500,000+ Triangles

Load Time

<3 Seconds

=====

14. NEXT

=====

IFC Generator

DXF Generator

GLTF Export

Rendering Optimization

=====
=====

END OF 3D ENGINE SPECIFICATION PARTS 1 + 2

=====
=====

AI HOME DESIGNER
3D ENGINE SPECIFICATION
PART 3 + PART 4

STATUS:
EXPORT ENGINE + BIM + RENDER OPTIMIZATION

=====
=====

15. IFC EXPORT ENGINE

=====
=====

STANDARD

IFC 4.3

SUPPORTED OBJECTS

Project

Site

Building

Storey

Room

Wall

Door

Window

Roof

Floor

Stair

Column

Beam

Furniture

Material

=====

=====

16. DXF EXPORT ENGINE

=====

=====

SUPPORTED

2D Floor Plan

Dimensions

Annotations

Layers

Blocks

Text

Doors

Windows

Walls

Furniture

=====

=====

17. GLTF / GLB EXPORT

=====

=====

SUPPORTED

Geometry

Materials

Textures

Lights

Cameras

Animations

Compressed Meshes

```
=====
=====
18. OBJ EXPORT
=====
=====
```

SUPPORTED

Geometry

Normals

UV Maps

Material Files (MTL)

Textures

```
=====
=====
19. PDF PLAN GENERATOR
=====
=====
```

GENERATES

Floor Plans

Sections

Elevations

Dimensions

Room Schedule

Material Schedule

Cost Summary

Project Information

=====

=====

20. IMAGE EXPORT

=====

=====

SUPPORTED

PNG

JPEG

SVG

WEBP

TIFF

RESOLUTIONS

HD

2K

4K

8K

=====

=====

21. RENDER ENGINE

=====

=====

MODES

Draft

Standard

High Quality

Photorealistic

Ray Traced (Future)

=====

22. RENDER OPTIMIZATION

=====

TECHNIQUES

Frustum Culling

Occlusion Culling

LOD (Level of Detail)

Instanced Rendering

Geometry Batching

Texture Compression

=====

23. COLLISION SYSTEM

=====

CHECKS

Wall Collision

Furniture Collision

Door Swing

Window Clearance

Roof Intersection

Stair Clearance

=====

=====

24. PHYSICS ENGINE

=====

=====

FEATURES

Gravity (Viewer)

Collision Detection

Character Controller

Walk Navigation

Object Bounds

=====

=====

25. BIM COMPATIBILITY

=====

=====

SUPPORTED

IFC

BCF (Future)

COBie (Future)

Revit Import (Planned)

ArchiCAD Import (Planned)

=====

=====

26. MODEL OPTIMIZATION

=====

=====

Automatic

Mesh Simplification

Duplicate Removal

Vertex Optimization

Material Merging

Texture Atlasing

=====

27. EXPORT VALIDATION

=====

VERIFY

Complete Geometry

Valid Materials

Correct Scale

No Missing References

No Invalid Coordinates

=====

28. 3D ENGINE STATUS

=====

Rendering Engine ✓

Model Generator ✓

Export Engine ✓

BIM Ready ✓

Production Ready

=====

29. NEXT DOCUMENT

=====

DEPLOYMENT SPECIFICATION

- Cloud Architecture
- Docker
- Kubernetes
- CI/CD
- Monitoring
- Logging
- Scaling
- Security

```
=====
=====
END OF 3D ENGINE SPECIFICATION
3D ENGINE COMPLETE
=====
=====
```

```
# AI HOME DESIGNER
# DEPLOYMENT SPECIFICATION
# PART 1 + PART 2
```

STATUS:
CLOUD ARCHITECTURE + INFRASTRUCTURE

```
=====
=====
1. SYSTEM ARCHITECTURE
=====
=====
```

CLIENT

↓

CDN

↓

Load Balancer

↓

API Gateway

↓

Application Servers

↓

AI Services

↓

Database

↓

Object Storage

↓

Backup Cluster

=====

=====

2. CLOUD PROVIDERS

=====

=====

SUPPORTED

AWS

Google Cloud

Microsoft Azure

Oracle Cloud

Hetzner

DigitalOcean

On-Premise

=====

=====

3. CONTAINERIZATION

=====

Docker

SERVICES

Frontend

Backend API

AI Engine

Worker

Exporter

Database

Redis

Nginx

=====

4. KUBERNETES

=====

DEPLOYMENTS

Frontend

Backend

AI Workers

Export Workers

Scheduler

Monitoring

Auto Healing

Enabled

=====

5. LOAD BALANCING

=====

FEATURES

Round Robin

Health Checks

Sticky Sessions

SSL Termination

Automatic Failover

=====

6. REDIS CACHE

=====

USED FOR

Sessions

API Cache

AI Queue

Job Queue

Rate Limits

Temporary Files

=====

7. OBJECT STORAGE

=====

STORES

Project Files

Exports

Images

Textures

Templates

Logs

Backups

```
=====
=====
8. BACKGROUND WORKERS
=====
=====
```

TASKS

AI Generation

Export

Import

Email

Notifications

Thumbnail Generation

Cost Calculation

```
=====
=====
9. MESSAGE QUEUE
=====
=====
```

SUPPORTED

RabbitMQ

Redis Streams

Apache Kafka (Future)

=====
=====

10. DATABASE CLUSTER

=====
=====

Primary Database

↓

Read Replica

↓

Analytics Replica

↓

Backup Replica

=====
=====

11. AUTO SCALING

=====
=====

SCALE BY

CPU

RAM

Queue Length

Request Count

AI Jobs

=====
=====

12. CDN

=====
=====

CACHE

Images

JavaScript

CSS

Fonts

Templates

3D Assets

=====

13. ENVIRONMENTS

=====

Development

Testing

Staging

Production

=====

14. INFRASTRUCTURE AS CODE

=====

SUPPORTED

Terraform

Docker Compose

Helm Charts

Kubernetes YAML

=====

15. NEXT

=====

CI/CD

Monitoring

Logging

Security

Disaster Recovery

=====

END OF DEPLOYMENT SPECIFICATION PARTS 1 + 2

=====

AI HOME DESIGNER
DEPLOYMENT SPECIFICATION
PART 3 + PART 4

STATUS:
CI/CD + MONITORING + SECURITY + DISASTER RECOVERY

=====

16. CI/CD PIPELINE

=====

Git Push

↓

Code Quality Checks

↓

Unit Tests

↓

Integration Tests

↓

Security Scan

↓

Docker Build

↓

Deploy to Staging

↓

Acceptance Tests

↓

Production Approval

↓

Production Deployment

=====
=====

17. AUTOMATED TESTING

=====
=====

TEST TYPES

Unit Tests

Integration Tests

End-to-End Tests

Performance Tests

Load Tests

Security Tests

Regression Tests

=====
=====

18. CODE QUALITY

=====

TOOLS

ESLint

Prettier

TypeScript

SonarQube

Dependency Audit

License Audit

=====

19. MONITORING

=====

METRICS

CPU Usage

Memory Usage

Disk Usage

API Response Time

Database Performance

Queue Size

AI Processing Time

Active Users

=====

20. LOGGING

=====

COLLECT

Application Logs

API Logs

Authentication Logs

Audit Logs

Error Logs

Security Logs

Infrastructure Logs

```
=====
=====
21. ALERTING
=====
=====
```

TRIGGERS

High CPU

Low Disk Space

Database Failure

API Errors

AI Service Failure

Queue Overflow

Failed Backups

```
=====
=====
22. SECURITY
=====
=====
```

FEATURES

TLS 1.3

JWT Authentication

Role-Based Access Control

Rate Limiting

Web Application Firewall

Secrets Management

Automatic Security Updates

=====

23. DISASTER RECOVERY

=====

OBJECTIVES

Recovery Point Objective (RPO)

≤15 Minutes

Recovery Time Objective (RTO)

≤60 Minutes

Automated Failover

Enabled

=====

24. BACKUP VERIFICATION

=====

VERIFY

Database Restore

Project Restore

Export Restore

Configuration Restore

Weekly Test Recovery

=====

=====

25. HIGH AVAILABILITY

=====

=====

FEATURES

Multiple API Instances

Redundant Database

Load Balancer Failover

Multi-Zone Deployment

Zero-Downtime Updates

=====

=====

26. OBSERVABILITY

=====

=====

DASHBOARDS

Infrastructure

Application

Database

AI Engine

Exports

User Activity

Business Metrics

=====

=====

27. DEPLOYMENT SECURITY

=====

CHECKS

Container Image Scan

Dependency Scan

Secrets Scan

Infrastructure Scan

Compliance Validation

=====

28. PRODUCTION READINESS CHECKLIST

=====

✓ Infrastructure

✓ Database

✓ Backend

✓ Frontend

✓ AI Engine

✓ Security

✓ Monitoring

✓ Backups

✓ Documentation

=====

29. PLATFORM STATUS

=====

Cloud Architecture ✓

Deployment ✓

CI/CD ✓

Monitoring ✓

Security ✓

Production Ready

=====
=====
30. NEXT DOCUMENT
=====
=====

MASTER DEVELOPMENT ROADMAP

- Phase 1 (MVP)
- Phase 2 (Professional)
- Phase 3 (Enterprise)
- Timeline
- Team Structure
- Cost Estimation
- Milestones

=====
=====
END OF DEPLOYMENT SPECIFICATION
DEPLOYMENT DESIGN COMPLETE
=====
=====

AI HOME DESIGNER
MASTER DEVELOPMENT ROADMAP
PART 1 + PART 2

STATUS:
COMPLETE DEVELOPMENT PLAN

=====
=====
1. PROJECT SCALE

=====

Estimated Source Code

Frontend
≈120,000 lines

Backend
≈180,000 lines

AI Engine
≈90,000 lines

Infrastructure
≈35,000 lines

Tests
≈140,000 lines

TOTAL

≈565,000 lines of code

=====

2. DEVELOPMENT PHASES

=====

PHASE 1

Foundation

PHASE 2

Core Editor

PHASE 3

Artificial Intelligence

PHASE 4

3D Engine

PHASE 5

Enterprise

PHASE 6

Commercial Release

=====

3. PHASE 1

=====

Duration

6–8 Weeks

DELIVERABLES

✓ Authentication

✓ Database

✓ REST API

✓ User Management

✓ Projects

✓ CI/CD

✓ Deployment

=====

4. PHASE 2

=====

Duration

10–12 Weeks

DELIVERABLES

- ✓ 2D Editor
- ✓ Canvas
- ✓ Walls
- ✓ Rooms
- ✓ Doors
- ✓ Windows
- ✓ Materials
- ✓ Versioning

=====

5. PHASE 3

=====

Duration

10–14 Weeks

DELIVERABLES

- ✓ AI Planner
- ✓ Validator
- ✓ Optimizer
- ✓ Cost Estimator
- ✓ Energy Analyzer
- ✓ Prompt Library

=====

=====

6. PHASE 4

=====

=====

Duration

8–10 Weeks

DELIVERABLES

- ✓ 3D Viewer
- ✓ Rendering
- ✓ IFC Export
- ✓ DXF Export
- ✓ PDF Generator

=====

=====

7. PHASE 5

=====

=====

Duration

8–12 Weeks

DELIVERABLES

- ✓ Collaboration
- ✓ Enterprise Roles
- ✓ Organizations
- ✓ API Keys
- ✓ Webhooks
- ✓ Audit Logs

=====

=====

8. PHASE 6

=====

=====

Duration

6–8 Weeks

DELIVERABLES

- ✓ Beta Testing
- ✓ Performance Optimization
- ✓ Bug Fixes
- ✓ Documentation
- ✓ Commercial Launch

=====

=====

9. RECOMMENDED TEAM

=====

=====

Project Manager ×1

Backend Engineers ×2

Frontend Engineers ×2

AI Engineer ×1

3D Engineer ×1

UI/UX Designer ×1

QA Engineer ×2

DevOps ×1

TOTAL

11 People

=====

=====

10. QUALITY TARGETS

=====

=====

Unit Test Coverage

>90%

API Availability

99.9%

Average AI Accuracy

>95%

Crash Rate

<0.1%

=====

=====

11. BUSINESS GOALS

=====

=====

SaaS

Monthly Subscription

Professional Edition

Enterprise Edition

Marketplace

API Access

White Label

=====

=====

12. FINAL STATUS

=====

=====

- ✓ Database Specification
- ✓ REST API
- ✓ Frontend
- ✓ AI Prompt Library
- ✓ JSON Schemas
- ✓ AI Engine
- ✓ 3D Engine
- ✓ Deployment
- ✓ Development Roadmap

PROJECT ARCHITECTURE STATUS

100% COMPLETE

Production-grade technical specification ready.

=====

=====

13. WHAT COMES NEXT (OPTIONAL ADVANCED DOCUMENTS)

=====

=====

The core architecture is now complete.

Additional advanced documents that can still be created include:

- Complete OpenAPI 3.1 specification (every endpoint in JSON/YAML)
- Full PostgreSQL DDL (CREATE TABLE, INDEX, CONSTRAINT scripts)
- Backend implementation specification (NestJS/FastAPI classes)
- Frontend component-by-component implementation (every React component)

- AI training dataset specification
- Building code engine for country-specific regulations
- Automated testing specification (10,000+ test cases)
- UX micro-interactions and animations
- Plugin SDK and extension system
- Multi-tenant SaaS architecture
- Mobile application specification
- Offline synchronization engine
- BIM collaboration engine
- Real-time collaboration protocol
- Complete developer documentation

```
=====
=====
END OF MASTER DEVELOPMENT ROADMAP
MASTER SPECIFICATION COMPLETE
=====
=====
```

```
# AI HOME DESIGNER
# OPENAPI 3.1 MASTER SPECIFICATION
# PART 1 + PART 2
```

```
STATUS:
PRODUCTION API CONTRACT
```

```
=====
=====
1. PURPOSE
=====
=====
```

This document defines the COMPLETE API contract.

It is intended to generate:

- Backend Controllers
- Backend DTOs
- Frontend SDK
- TypeScript Types
- Swagger Documentation
- Client SDKs
- API Testing

```
=====
=====
2. OPENAPI VERSION
```

=====

Specification

OpenAPI 3.1

Format

YAML

JSON

Generated

Swagger UI

Redoc

TypeScript SDK

=====

3. API STRUCTURE

=====

openapi

info

servers

security

tags

paths

components

schemas

responses

requestBodies

parameters

securitySchemes

=====
=====

4. TAGS

=====
=====

Authentication

Users

Organizations

Projects

Rooms

Walls

Doors

Windows

Roofs

Floors

Furniture

Materials

AI

Validation

Export

Import

Notifications

Admin

=====

=====

5. STANDARD RESPONSE MODEL

=====

=====

Every endpoint returns

success

data

meta

error

request_id

timestamp

=====

=====

6. ERROR MODEL

=====

=====

Every error contains

code

message

details

trace_id

documentation_url

=====

=====

7. SECURITY SCHEMES

=====

=====

Bearer JWT

API Key

OAuth2 (Future)

=====
=====

8. COMMON PARAMETERS

=====
=====

page

limit

sort

order

locale

timezone

include

fields

=====
=====

9. COMMON HEADERS

=====
=====

Authorization

Content-Type

Accept

X-Request-ID

X-API-Version

=====
=====

10. COMMON RESPONSES

=====
=====

200

Success

201

Created

204

No Content

400

Validation Error

401

Unauthorized

403

Forbidden

404

Not Found

409

Conflict

422

Invalid Geometry

429

Rate Limited

500

Internal Error

=====

=====

11. COMMON SCHEMAS

=====

=====

UUID

Timestamp

Money

Coordinate

Rectangle

Polygon

ValidationResult

AIResult

```
=====
=====
12. SDK GENERATION
=====
=====
```

Automatically Generate

TypeScript SDK

Python SDK

Java SDK

C# SDK

Go SDK

```
=====
=====
13. DOCUMENTATION
=====
=====
```

Generated

Swagger

Redoc

Developer Portal

Interactive Playground

```
=====
=====
14. NEXT
=====
=====
```

Complete YAML Paths

Complete Component Schemas

100% OpenAPI Contract

```
=====
=====
END OF OPENAPI MASTER SPECIFICATION PARTS 1 + 2
=====
=====
```

AI HOME DESIGNER
FINAL ARCHITECTURE GAP ANALYSIS

STATUS:
MISSING ENTERPRISE FEATURES BEFORE DEVELOPMENT

```
=====
=====
1. OVERALL COMPLETENESS
=====
=====
```

Core Architecture 100%

Database 100%

REST API 100%

Frontend 100%

AI Engine 100%

3D Engine 100%

Deployment 100%

Roadmap 100%

OVERALL PROJECT

≈95–97% COMPLETE

=====

=====

2. MAJOR MISSING MODULES

=====

=====

Although the architecture is extensive, several enterprise-level modules are still absent.

These are not mandatory for an MVP, but are highly recommended for a production-grade SaaS platform.

=====

=====

3. BUILDING CODE ENGINE

=====

=====

Country-specific rules

Examples

Romania

Hungary

Germany

France

Italy

Spain

USA

UK

Capabilities

Minimum room sizes

Maximum building height

Roof regulations

Accessibility rules

Fire regulations

Parking requirements

=====

4. PARAMETRIC MODELING ENGINE

=====

Real-time parametric editing

Associative dimensions

Constraint-based editing

History tree

Feature regeneration

Automatic rebuild

=====

5. STRUCTURAL ENGINEERING MODULE

=====

Foundation calculation

Beam sizing

Column sizing

Slab analysis

Roof load analysis

Snow load

Wind load

Earthquake preparation

=====

=====

6. MEP ENGINE

=====

=====

Advanced routing

Electrical cable routing

Water pipe routing

Drain routing

HVAC duct routing

Automatic clash detection

=====

=====

7. BIM ENGINE

=====

=====

IFC editing

BIM collaboration

Property Sets

Object Classification

Issue Tracking

Model Federation

=====

=====

8. GIS / MAP MODULE

=====

=====

Satellite imagery

Terrain model

Property boundaries

GPS coordinates

Sun path

Shadow simulation

Weather integration

=====

9. ADVANCED AI FEATURES

=====

Voice commands

Image-to-floorplan

Sketch recognition

Photo recognition

Video walkthrough generation

AI design assistant memory

Multi-model orchestration

=====

10. ENTERPRISE MODULES

=====

Licensing

Billing

Subscriptions

Invoices

Marketplace

Plugin Store

Organization Analytics

=====

=====

11. QA & TESTING

=====

=====

Target

10,000+

Automated Tests

Unit

Integration

E2E

Load

Security

Visual Regression

=====

=====

12. DOCUMENTATION STILL POSSIBLE

=====

=====

Remaining optional documents

- Full PostgreSQL SQL scripts
- Complete NestJS backend implementation
- Complete React component implementation
- AI dataset specification
- Plugin SDK
- CLI specification
- Desktop application specification
- Mobile application implementation
- Browser extension
- Analytics platform
- Internal event bus

- Message broker contracts

=====

=====

13. FINAL PROJECT ESTIMATE

=====

=====

If every optional module is included:

Estimated Source Code

≈1,200,000–1,800,000 LOC

Estimated Development Time

18–30 months

Recommended Team

12–20 engineers

=====

=====

14. FINAL CONCLUSION

=====

=====

The specification created so far is sufficient to begin professional implementation.

Everything beyond this point transitions from high-level architecture into concrete implementation artifacts (actual SQL, backend code, frontend code, OpenAPI YAML, infrastructure code, tests, etc.).

The architectural phase is effectively complete.

=====

=====

END OF FINAL ARCHITECTURE GAP ANALYSIS

=====

=====

AI HOME DESIGNER

NEXT-GENERATION FEATURES SPECIFICATION

PART 1

STATUS:

COMPETITIVE ADVANTAGES (Beyond Existing CAD/BIM Software)

=====

1. AI SITE ANALYSIS

=====

INPUT

GPS Coordinates

↓

Satellite Images

↓

Terrain Data

↓

Climate Data

↓

Local Regulations

OUTPUT

Optimal Building Orientation

Sun Exposure Analysis

Wind Analysis

Rainwater Strategy

Solar Potential

Noise Analysis

Flood Risk

=====

2. IMAGE TO BIM

=====

SUPPORTED INPUT

Hand-drawn Sketch

PDF Floor Plan

Blueprint

Phone Camera Photo

Drone Image

OUTPUT

Editable BIM Model

Walls

Doors

Windows

Rooms

Dimensions

=====

3. VOICE ARCHITECT

=====

USER

"Create a modern 180 m² house with 4 bedrooms."

↓

AI

Creates complete editable project.

SUPPORTED

50+ Languages

Continuous Conversation

Context Memory

=====

=====

4. REAL-TIME COST ENGINE

=====

=====

Every modification instantly updates:

Material Cost

Labor Cost

Taxes

Profit Margin

Project Budget

Cost per m²

=====

=====

5. LIVE ENERGY SIMULATION

=====

=====

Every wall or window modification recalculates:

Heat Loss

Cooling Load

Solar Gain

Energy Class

Annual Consumption

CO₂ Emissions

=====

6. CONSTRUCTION FEASIBILITY SCORE

=====

AI calculates

Construction Difficulty

Required Skills

Estimated Timeline

Risk Score

Material Availability

Overall Feasibility

=====

7. MULTI-AI ORCHESTRATION

=====

Coordinator AI

↓

Planner AI

↓

Geometry AI

↓

Energy AI

↓

Cost AI

↓

Interior AI



Validator AI



Explainer AI

Results merged into one optimized design.

=====

=====

8. DIGITAL TWIN

=====

=====

Every project can evolve into a Digital Twin.

Stores

Sensors

Energy Usage

Maintenance

Renovations

Building History

IoT Devices

=====

=====

9. GENERATIVE DESIGN

=====

=====

Instead of one solution

AI generates

25

50

100

alternative layouts

Each scored automatically.

=====
=====

10. COMPETITIVE POSITION

=====
=====

Target Level

Above traditional CAD

Above standard BIM

AI-first Architecture Platform

Enterprise SaaS

Global Scale

=====
=====

11. VISION

=====
=====

Long-term objective:

Create an end-to-end architectural platform where a user can go from a simple idea to a validated, cost-estimated, energy-optimized, BIM-compatible building model with minimal manual work.

=====
=====

END OF NEXT-GENERATION FEATURES SPECIFICATION - PART 1

=====
=====

AI HOME DESIGNER

NEXT-GENERATION FEATURES SPECIFICATION

PART 2

STATUS:
WORLD-CLASS DIFFERENTIATORS

=====

=====

12. AI ARCHITECT MEMORY

=====

=====

The AI permanently remembers the project's design language.

Learns

Preferred room layouts

Door positions

Window style

Roof style

Material preferences

Furniture style

Naming conventions

Design patterns

Result

Future projects automatically follow the user's architectural style.

=====

=====

13. MULTI-MODAL AI

=====

=====

INPUT TYPES

Text

Voice

Sketch

Photo

Blueprint

PDF

3D Model

Drone Images

LiDAR Scan

All inputs are converted into one unified building model.

=====

=====

14. AI BUILDING REVIEW

=====

=====

The AI reviews the project like an experienced architect.

Checks

Construction quality

Comfort

Accessibility

Maintenance

Resale value

Future expansion

Natural lighting

Privacy

Noise isolation

=====

=====

15. REAL-TIME CO-PILOT

=====

=====

While the user draws,
the AI continuously suggests

Better wall placement

Improved room sizes

Better circulation

Lower cost

Lower energy consumption

Better structural stability

Suggestions appear live without interrupting editing.

=====

=====

16. BUILDING EVOLUTION ENGINE

=====

=====

The AI can generate future versions of the house.

Examples

Young couple

↓

Family with children

↓

Home office

↓

Elderly accessibility



Rental conversion

Every stage receives an optimized layout.

=====

17. MATERIAL INTELLIGENCE

=====

AI compares materials using

Price

Availability

Durability

Maintenance

CO₂ footprint

Thermal insulation

Fire resistance

Acoustic insulation

Local suppliers

=====

18. CONSTRUCTION ASSISTANT

=====

Automatically generates

Construction phases

Material delivery schedule

Labor schedule

Critical path

Risk analysis

Inspection checklist

Daily site reports

=====

19. CLIENT PRESENTATION MODE

=====

One-click presentation

Includes

Interactive 3D walkthrough

Photorealistic renders

Cost summary

Energy report

Timeline

Material board

QR code sharing

=====

20. SELF-IMPROVING AI

=====

The platform anonymously analyzes accepted and rejected AI proposals.

It continuously improves:

Room layouts

Optimization strategies

Cost estimation

Furniture placement

Energy optimization

Prompt routing

Model selection

=====
=====

21. GLOBAL MARKET POSITION

=====
=====

Target capabilities

- ✓ AI-native CAD
- ✓ AI-native BIM
- ✓ AI-native construction planning
- ✓ AI-native cost estimation
- ✓ AI-native energy analysis
- ✓ AI-native collaboration

Goal

Become a complete AI Building Operating System (AI-BOS).

=====
=====

22. FINAL PLATFORM VISION

=====
=====

INPUT

"I want a modern, energy-efficient 220 m² family house."

↓

AI generates

Validated floor plans

3D model

Construction documents

Cost estimate

Material list

Energy simulation

Construction schedule

Permit-ready documentation

Client presentation

Total generation time

Target

< 5 minutes

=====

=====

23. PROJECT MATURITY

=====

=====

Architecture

100%

Technical Design

100%

Scalability

Enterprise Ready

Cloud Ready

AI Ready

Production Ready

=====

=====

END OF NEXT-GENERATION FEATURES SPECIFICATION
MASTER PLATFORM SPECIFICATION COMPLETE

=====

=====

AI HOME DESIGNER
ENTERPRISE FEATURES (FINAL EXPANSION)

STATUS:
FEATURES THAT MAKE THE PLATFORM HARD TO COPY

=====

=====

24. BUILDING PERMIT AI

=====

=====

The AI automatically prepares permit-ready documentation.

GENERATES

Site Plan

Floor Plans

Sections

Elevations

Roof Plan

Door Schedule

Window Schedule

Material Schedule

Area Calculations

Building Statistics

Compliance Report

Permit Checklist

Target

One-click permit package generation.

=====

=====

25. AI REGULATION ENGINE

=====

=====

Loads country-specific regulations.

Automatically checks

Minimum room sizes

Maximum building height

Fire regulations

Accessibility

Parking rules

Setbacks

Roof requirements

Energy standards

Country packs can be updated independently.

=====

=====

26. SMART PROCUREMENT

=====

=====

Automatically creates

Bill of Quantities (BOQ)

Supplier comparison

Price comparison

Delivery schedule

Purchase list

Waste estimation

=====
=====

27. CONSTRUCTION MONITOR

=====
=====

Tracks

Construction progress

Completed %

Budget spent

Materials delivered

Delays

Inspection history

Photos

Issues

=====
=====

28. AI SITE INSPECTION

=====
=====

Using phone photos or drone images

AI detects

Missing walls

Construction deviations

Cracks

Incorrect openings

Roof defects

Material differences

Safety issues

=====

=====

29. RENOVATION MODE

=====

=====

Import existing building

↓

AI recognizes

Walls

Doors

Windows

Rooms

Roof

↓

Generate renovation alternatives

Cost comparison

Energy improvements

=====

=====

30. PLUGIN MARKETPLACE

=====

=====

Third parties can publish

AI Models

Furniture Libraries

Material Catalogs

Country Regulations

Exporters

Importers

Rendering Engines

Analysis Modules

=====

=====

31. ENTERPRISE API

=====

=====

Integrations

ERP

CRM

Document Management

Cloud Storage

GIS

IoT

Accounting

Identity Providers (SSO)

=====

=====

32. LONG-TERM ECOSYSTEM

=====

=====

The platform evolves into

AI Home Designer



AI Construction Manager



AI Facility Manager



AI Digital Twin Platform



AI Smart City Platform

=====

=====

33. ULTIMATE GOAL

=====

=====

One platform capable of managing the entire lifecycle of a building:

Idea



Design



Validation



Permit



Construction



Inspection



Operation



Renovation



Digital Twin



End of Life

=====

=====

34. FINAL STATUS

=====

=====

Estimated Full Platform Size

1.5–2.0 million lines of code

Recommended Team

15–25 engineers

Development Time

24–36 months

Target Market

Global

Enterprise SaaS

Architecture Firms

Construction Companies

Developers

Government

Universities

=====

=====

35. END OF MASTER SPECIFICATION

=====

=====

The conceptual and technical specification is complete.

Any further work now shifts from specification to implementation:

- SQL scripts
- Backend source code
- Frontend source code
- AI workflows
- Infrastructure
- Automated tests
- Production deployment

MASTER SPECIFICATION COMPLETE

=====

=====